Final Report

Food Ordering Behaviour and Consumer Trends: A Structured Analysis of Choices and Habits

Abstract

The Food Ordering Behaviour and Consumer Trends project is an interactive business intelligence solution built with Tableau and delivered through a Flask web application. The project analyzes order demand, cuisine and meal-type preferences, city-wise trends, and delivery performance using a food ordering dataset. Through an interactive dashboard and story, the project converts raw order data into visual insights that support Business Analysts, Operations Managers, and customers alike.

Objectives

- Analyze customer ordering behaviour across cuisines, meal types, and cities.
- Monitor business performance using KPIs (Total Orders, Total Order Value, Average Delivery Time).
- Evaluate delivery efficiency through delivery time distribution.
- Study the relationship between order value and delivery fee.
- Support informed decision-making using an interactive dashboard and story.
- Deliver the analysis through a responsive, web-accessible interface.

Scenarios Addressed

Rahul Sharma - Business Analyst

Analyzes order trends, cuisine preferences, and city-wise demand using the Tableau dashboard, studying Order Value and Meal Type to uncover insights that support strategic, data-driven business decisions and track demand across time periods.

Priya Mehta - Operations Manager

Focuses on delivery performance and operational efficiency, analyzing delivery time distribution and order volume by city to optimize resource allocation, monitor peak hours, and reduce service gaps.

Neha Reddy - Customer

Values quick delivery, reasonable pricing, and cuisine variety, ordering mostly during lunch and dinner. Her satisfaction - and likelihood to reorder - depends heavily on timely delivery and overall service quality.

Methodology

1. Data Collection & Extraction from the source dataset
2. Data Cleaning and Preparation
3. KPI and Calculated Field Development
4. Dashboard Design
5. Story Creation
6. Performance Testing
7. Web Integration with Flask

8 Documentation

Tools Used

- Tableau Desktop / Tableau Public
- Microsoft Excel
- Python 3 with Flask
- HTML5, CSS3, JavaScript
- Windows Operating System

Dashboard Features

- Total Orders KPI
- Total Order Value KPI
- Average Delivery Time KPI
- Cuisine Analysis (Bar Chart)
- Meal Type Distribution (Pie / Donut Chart)
- City-wise Demand (Map / Bar Chart)
- Delivery Time Distribution (Histogram)
- Order Value vs Delivery Fee (Scatter Plot)
- Interactive Filters
- Tableau Story

Web Integration

The published Tableau Dashboard and Story are embedded into a Flask web application using the tableau-viz web component. The application provides a Home page (project overview and KPIs), a Dashboard page, a Story page, and an About page, all built with a responsive HTML/CSS/JS front end.

Results

The dashboard successfully identifies ordering patterns, top-performing cities and cuisines, peak mealtimes, and delivery time distribution. Interactive filters and the Tableau Story enable Business Analysts, Operations Managers, and customers to explore the data and reach conclusions quickly.

Future Enhancements

- Real-time / streaming data integration.
- Predictive analytics for demand forecasting using machine learning.
- Deeper customer segmentation.
- Personalized cuisine and restaurant recommendations.
- Integration with a cloud database for live dashboard refresh.
- User authentication and role-based dashboard views in the Flask app.

Conclusion

The Food Ordering Behaviour and Consumer Trends project provides an efficient, interactive platform for analyzing food ordering data. By combining Tableau analytics with a Flask-based web interface, the project simplifies complex ordering and delivery data into actionable visual insights, helping the business improve operational efficiency, delivery speed, and overall customer satisfaction.